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[22EEDE541]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

NON-CONVENTIONAL ENERGY SOURCES

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Justify the need of renewable energy sources in present scenario.	**L1**	**CO1**	**02**
	b.	What is the principle of solar photovoltaic?	**L1**	**CO2**	**02**
	c.	Why horizontal axis wind turbines are preferred over vertical axis wind turbines?	**L1**	**CO3**	**02**
	d.	Differentiate between open cycle and closed cycle OTEC based power generation.	**L1**	**CO4**	**02**
	e.	What are the benefits of hydrogen energy?	**L1**	**CO5**	**02**
		PART-B	**6X5=30**		
2.	**a.**	Explain the environmental aspects of energy utilization.	**L2**	**CO1**	**05**
		OR			
	b.	List the merits and demerits of conventional sources of energy?	**L1**	**CO1**	**05**
3.	**a.**	Explain solar radiation on the earth surface with neat schematic diagram.	**L2**	**CO2**	**05**
		OR			
	b.	Explain concentrating and non-concentrating types of solar collectors.	**L2**	**CO2**	**05**
4.	**a.**	Illustrate with neat diagram the working of vertical axis wind turbine and explain its components.	**L2**	**CO3**	**05**
		OR			
	b.	Explain with block diagram of wind energy conversion devices.	**L2**	**CO6**	**05**
5.	**a.**	Describe with simple line diagram single basin tidal power plant.	**L2**	**CO4**	**05**
		OR			
	b.	Explain the floating type dome plants with schematic diagram.	**L2**	**CO4**	**05**
6.	**a.**	What is the principle of MHD power generation and discuss about the main parts of an MHD generator?	**L2**	**CO5**	**05**
		OR			
	b.	Illustrate the methods of hydrogen production and transportation.	**L2**	**CO5**	**05**
7.	**a.**	What is wave energy? What are the various devices used in harnessing this energy?	**L1**	**CO2**	**05**
		OR			
	b.	Explain briefly the open Ocean Thermal Energy Conversion (OTEC) cycles with a neat diagram.	**L2**	**CO4**	**05**

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		PART-C	**6X10=60**		
8.	**a.**	What are conventional and non-conventional energy source with an examples? List atleast five States where coal deposits are concentrated in India.	**L1**	**CO1**	**10**
		OR			
	b.	Explain Importance of Energy Consumption as Measure of Prosperity.	**L2**	**CO1**	**10**
9.	**a.**	Explain the working of solar cooker with a neat diagram	**L2**	**CO2**	**10**
		OR			
	b.	With the help of schematic diagram, Discuss the measurement of solar radiations.	**L2**	**CO2**	**10**
10.	**a.**	Illustrate the safety and environmental aspects in wind energy system.	**L2**	**CO3**	**10**
		OR			
	b.	State the classification of wind energy system. Explain offshore wind energy.	**L2**	**CO3**	**10**
11.	**a.**	What is biomass energy? Explain biogas plant with neat sketch.	**L2**	**CO4**	**10**
		OR			
	b.	Discuss the working principles and processes involved in the Geothermal Energy conversion.	**L2**	**CO4**	**10**
12.	**a.**	Illustrate the hydrogen storage technique.	**L2**	**CO5**	**10**
		OR			
	b.	Discuss the technologies behind mini-micro hydropower plants.	**L2**	**CO6**	**10**
13.	**a.**	Explain Hydrogen generation process with a neat diagram.	**L2**	**CO5**	**10**
		OR			
	b.	Explain Photovoltaic system components and different applications.	**L2**	**CO2**	**10**